

KRpep-Zero study brief

A reproducible study of chemistry and control validation

Main finding. Full-molecule encoding corrected the measured chemical-geometry failures in the capped KRpep controls. It did not yield sufficient confidence separation or reliable experimental-pose recovery to support selecting novel binders. The independent Protenix diagnostic positive control recovered 0/41 native contacts in each seed (17, 42 and 101); scramble contact recovery is NA, and chemistry checks failed.

Evidence	Observed outcome
Reference	GDP-bound KRAS4B G12D 1-169 and KRpep-2d, PDB 5XCO
Actual model work	ESMC profiles; 2 rejected generator pilot structures; 18 control predictions total (12 Boltz + 6 Protenix)
Control identities	KRpep-2d positive and proposed scramble; runs are model samples, not biological replicates
Revision 2 chemistry	6/6 outputs: all 183 bonds in bounds and all 20 stereocenters correct
Revision 2 selection	Gap +0.0332 versus required +0.150; gate failed
Protenix base diagnostic	6/6 outputs; positive 0/41 each seed (17, 42, 101); scramble NA; RMSD 29.06 / 30.93 / 30.58 Å; chemistry checks failed
Independent experiments	None performed in this campaign

Available analyses

Use the all-18 table's View pose for one selected row. In the viewer set Model / representation to reference, v1, v2 or protenix, then choose Control and Seed; the UI shows one selected prediction plus the reference at a time. Use Reference peptide, Predicted target, Target surface and Prediction focus, then change Seed to compare runs. Inspect contact recovery and RMSD alongside the selected pose. The package contains source data, exact atom mappings, scripts, declared protocols and an offline reproduction receipt. The 18 predictions represent two control identities across model runs, not 18 independent peptides or biological replicates.

What remains unresolved

The scramble is not a demonstrated nonbinder. Published weak-binding analogs offer a better-supported future benchmark, but their modified chemistry must be verified first. No novel peptide, mutant selectivity, developability or therapeutic activity is established.

Discussion focus: whether the proposed benchmark and pose metrics are sufficient for a subsequent candidate-selection study. The full report includes methods, all-seed figures, experimental control candidates and references.